

ABSTRACT

The rapid increase in population, urbanization, and industrialization has led to a significant rise in waste generation, especially along highways and public areas. Traditional waste management systems mainly depend on manual waste collection and segregation, which are inefficient, time-consuming, and hazardous to human health. To address these challenges, this project presents a **Smart Trash Bot: Autonomous Waste Collection and Segregation System with GSM Module** using embedded systems and IoT-based technologies.

The proposed system is designed using a dual Arduino-based architecture integrated with ultrasonic sensors, wet sensors, servo motors, DC motors, GSM module, LCD display, and a robotic arm mechanism. The ultrasonic sensor is used to detect waste objects and monitor the fill level of waste bins, while the wet sensor identifies the type of waste based on moisture content. The robotic arm automatically picks and segregates waste into wet and dry categories using servo motor control.

The system operates autonomously by continuously moving along roadside or public environments, detecting waste, collecting it through the robotic arm, and placing it into appropriate bins. Real-time monitoring of bin levels is achieved using ultrasonic sensors placed inside the bins. When the bins become full, the GSM module automatically sends alert messages to the concerned authorities, ensuring timely waste disposal and preventing overflow. The LCD display provides real-time status updates and system messages, while the buzzer generates audio alerts during operation.

The proposed system reduces manual effort, improves waste handling efficiency, enhances public hygiene, and minimizes environmental pollution. It is cost-effective, scalable, and suitable for smart cities, highways, industrial areas, railway stations, and other public environments. By integrating automation, robotics, and communication technologies, the system provides an intelligent and efficient solution for modern waste management applications.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO SMART WASTE MANAGEMENT SYSTEMS

The rapid growth of population, urbanization, and industrialization has resulted in a substantial increase in the generation of solid waste across the globe. With expanding cities and increasing consumption patterns, the amount of waste produced daily has reached critical levels. Efficient waste management has therefore become one of the most pressing challenges faced by modern society. Traditional waste management systems, which rely on manual collection, fixed schedules, and minimal monitoring, are no longer sufficient to handle the growing volume and complexity of waste.

Conventional methods of waste management often involve collecting mixed waste from bins and transporting it to landfills or dumping sites without proper segregation. This leads to several environmental issues such as air pollution caused by the burning of waste, water contamination due to leachate, and soil degradation. Moreover, the lack of efficient monitoring results in overflowing bins, unpleasant odors, and the spread of diseases, negatively impacting public health and hygiene.

To overcome these limitations, the concept of Smart Waste Management Systems has been introduced. These systems utilize advanced technologies such as the Internet of Things (IoT), sensors, cloud computing, and automation to improve the efficiency and effectiveness of waste management processes. IoT enables communication between devices through the internet, allowing real-time data collection, monitoring, and control.

In a smart waste management system, sensors are embedded in waste bins to monitor various parameters. Ultrasonic sensors measure the fill level of the bin, moisture sensors help differentiate between wet and dry waste, gas sensors detect harmful emissions, and weight sensors measure the amount of waste collected. These sensors are connected to a microcontroller, such as Arduino or NodeMCU, which processes the data and transmits it to a centralized system or cloud platform via communication technologies like Wi-Fi or GSM.

The integration of cloud computing allows data to be stored, analyzed, and accessed remotely through web or mobile applications. This enables authorities to monitor the status of waste bins in real time and take appropriate actions when necessary. For example, when a bin reaches its maximum capacity, the system can automatically send alerts to waste collection personnel, ensuring timely disposal and preventing overflow.

Another important feature of smart waste management systems is automatic waste segregation. Segregation at the source is essential for effective recycling and disposal. In traditional systems, segregation is performed manually, which is inefficient, time-consuming, and hazardous. Smart systems use sensors or image processing techniques to classify waste into categories such as biodegradable (organic waste), non-biodegradable (plastics, metals), and recyclable materials. This improves recycling efficiency and reduces the environmental impact of waste.

Additionally, smart waste management systems contribute to operational efficiency by enabling optimized waste collection routes. Instead of following fixed schedules, waste collection vehicles can be directed based on real-time data, reducing fuel consumption, operational costs, and carbon emissions. This approach not only improves efficiency but also supports sustainable development.

The implementation of smart waste management systems is a key component of smart city initiatives, where technology is used to improve infrastructure, resource management, and quality of life. By ensuring proper waste handling, these systems help create cleaner, healthier, and more sustainable urban environments.

Despite the numerous benefits, smart waste management systems also face certain challenges, including high initial investment costs, maintenance requirements, dependence on reliable internet connectivity, and potential issues related to sensor accuracy and data security. However, with continuous advancements in technology and increasing awareness about environmental sustainability, these challenges can be effectively addressed.

In conclusion, smart waste management systems represent a modern and efficient solution to the growing problem of waste management. By integrating IoT, sensors, and automation, these systems enhance efficiency, reduce environmental impact, and contribute to the development of smart and sustainable cities.



Fig.1.1 Smart Waste Management System Overview

1.2 NEED FOR HIGHWAY WASTE MANAGEMENT SYSTEM

Highways are essential components of a country's transportation infrastructure, facilitating the movement of people and goods over long distances. They connect urban and rural areas, support economic activities, and play a vital role in national development. However, waste management along highways is often neglected, leading to serious environmental, health, and aesthetic issues.

Travelers frequently dispose of waste such as plastic bottles, food wrappers, cans, and other materials along roadsides. In the absence of proper waste disposal facilities, this waste accumulates over time, creating unsightly and unhygienic conditions. The problem is further aggravated by the lack of regular monitoring and collection mechanisms.

Accumulated waste along highways can have several negative impacts. It can block drainage systems, leading to water stagnation and increasing the risk of mosquito breeding and the spread of diseases. It also poses a threat to wildlife, as animals may consume plastic and other harmful materials. Additionally, waste accumulation contributes to soil and water pollution, affecting the surrounding environment.

From an aesthetic perspective, littered highways create a negative impression on travelers and tourists, reducing the overall appeal of the region. This can impact tourism and economic growth. Moreover, manual cleaning of highways is labor-intensive, time-consuming, and often inefficient.

To address these challenges, there is a need for an efficient and intelligent waste management system specifically designed for highway environments. A smart highway waste management system uses IoT-based smart bins placed at

regular intervals along highways, toll plazas, and rest areas. These bins are equipped with sensors to monitor waste levels in real time.

When the bins reach a certain threshold, alerts are sent to waste management authorities, enabling timely collection and preventing overflow. This ensures cleanliness, reduces environmental impact, and improves the overall travel experience.

Furthermore, the integration of smart systems can help optimize waste collection routes, reducing fuel consumption and operational costs. By maintaining clean and hygienic highways, such systems contribute to environmental protection, public health, and sustainable development.



Fig.1.2 Waste Accumulation on Highways

1.3 PROBLEM STATEMENT

The existing waste management system, particularly in highway areas, faces numerous challenges that affect its efficiency and effectiveness. One of the major problems is the lack of proper waste disposal infrastructure, which leads to the accumulation of waste along roadsides.

There is no real-time monitoring system to track the status of waste bins, resulting in overflow and unhygienic conditions. Waste collection is typically carried out based on fixed schedules rather than actual demand, leading to inefficient utilization of resources.

Another significant issue is the absence of proper waste segregation. Mixed waste reduces recycling efficiency and increases environmental pollution.

Manual handling of waste exposes workers to hazardous materials, posing serious health risks.

Additionally, poor waste management practices contribute to environmental degradation, including air, water, and soil pollution. Waste accumulation along highways also affects the aesthetic value of the surroundings and creates inconvenience for travelers.

These challenges highlight the need for a smart, automated, and efficient waste management system that can address the limitations of traditional methods and ensure effective waste handling.

1.4 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop an IoT-based smart waste management system for highways.

The specific objectives include:

- To develop a smart waste bin system equipped with sensors
- To monitor waste levels in real time using IoT technology
- To implement automatic waste segregation mechanisms
- To generate alerts when bins are full
- To reduce manual effort and improve worker safety
- To maintain cleanliness and hygiene along highways
- To optimize waste collection processes
- To promote environmentally sustainable waste management practices

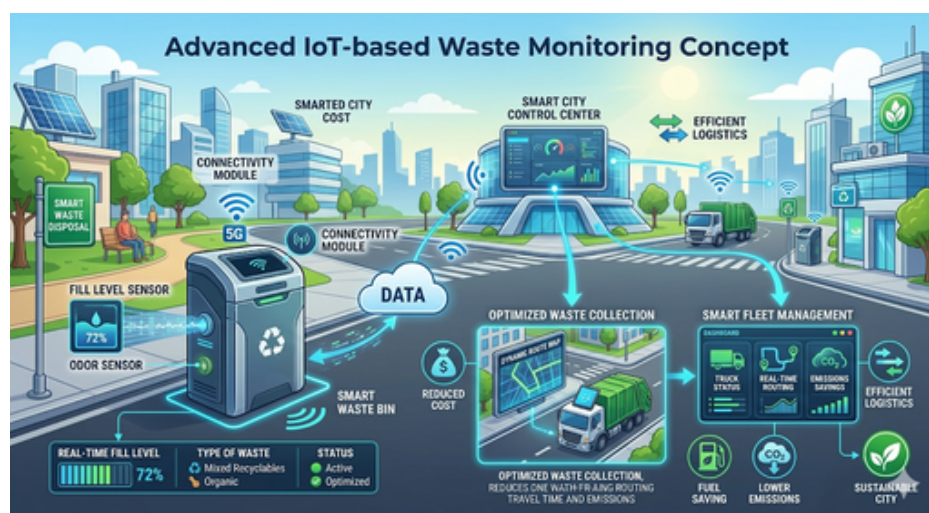


Fig.1.3 Iot Based waste monitoring concept

1.5 SCOPE OF THE PROJECT

The scope of this project involves the design and implementation of a smart waste management system using IoT technology for highway applications.

The project includes:

- Deployment of sensor-based smart bins
- Integration of microcontrollers and communication modules
- Real-time monitoring using cloud platforms
- Basic automation for waste segregation
- Alert system for efficient waste collection

The system can be applied in:

- Highways and expressways
- Toll plazas and rest areas
- Public transportation routes

However, the project is limited to a prototype-level implementation. Large-scale deployment, advanced AI-based waste classification, and full integration with municipal systems are beyond the scope of this study.

CHAPTER 2 LITERATURE SURVEY

2.1 INTRODUCTION

Waste management has become a critical global concern due to the rapid increase in population, urbanization, and industrialization. The traditional methods of waste collection and disposal are no longer sufficient to handle the growing volume of waste generated daily. Inefficient waste handling leads to environmental pollution, health hazards, and resource wastage.

In recent years, researchers have focused on developing smart waste management systems using advanced technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), machine learning, and automation. These technologies aim to improve waste segregation, enable real-time monitoring, and optimize waste collection processes.

This chapter presents a comprehensive review of existing research works related to smart waste monitoring and segregation systems. It highlights various techniques, technologies, advantages, and limitations of previously proposed systems. A comparative analysis is also provided to identify research gaps and justify the need for the proposed system.

2.2 REVIEW OF EXISTING LITERATURE

2.2.1 IOT-BASED SMART WASTE MONITORING SYSTEMS

IoT-based systems are among the most widely researched solutions in waste management. These systems use sensors, communication modules, and cloud platforms to monitor waste levels in real time.

Patel et al. (2020) proposed a sensor-based waste monitoring system that uses ultrasonic sensors to detect the fill level of waste bins. The collected data is transmitted to a cloud server using IoT technology, allowing authorities to monitor bin status remotely. The system generates alerts when bins are full, enabling timely waste collection.

Singh and Yadav (2021) developed an IoT-based garbage monitoring system that integrates sensors with a microcontroller and a mobile application. The system improves efficiency by reducing unnecessary waste collection trips and optimizing routes.

Bansal and Sharma (2022) presented an advanced IoT-based waste monitoring system that focuses on smart city applications. Their system integrates cloud computing and real-time data analytics to improve decision-making and operational efficiency.

KEY FEATURES:

- Real-time monitoring of waste levels
- Automated alert system
- Remote access through mobile/web applications
- Integration with cloud platforms

ADVANTAGES:

- Prevents overflow of bins
- Reduces operational cost
- Improves efficiency of waste collection

LIMITATIONS:

- Dependence on internet connectivity
- Requires regular maintenance of sensors



Fig.2.2 Autonomous Cleaning Robot

2.2.2 SENSOR-BASED WASTE SEGREGATION SYSTEMS

Waste segregation is an essential step in effective waste management. Several researchers have proposed sensor-based systems for automatic segregation of waste.

These systems typically use:

- Moisture sensors → to identify wet waste
- Infrared sensors → to detect dry waste
- Inductive sensors → to detect metal waste

Patel et al. (2020) also proposed a system that combines waste monitoring with basic segregation techniques. The system separates waste into biodegradable and non-biodegradable categories using sensor inputs.

Other researchers have highlighted that automated segregation reduces manual labor and improves recycling efficiency. It also minimizes health risks associated with handling hazardous waste.

ADVANTAGES:

- Reduces human intervention
- Improves accuracy in segregation
- Enhances recycling efficiency

LIMITATIONS:

- Limited classification capability
- Sensor errors may affect accuracy

2.2.3 IOT-BASED WASTE SEGREGATION AND MONITORING SYSTEMS

Some studies combine both monitoring and segregation functionalities into a

single system.

Singh and Yadav (2021) proposed a hybrid system that uses sensors for segregation and IoT for monitoring. The system automatically classifies waste and sends real-time data to a cloud platform.

Bansal and Sharma (2022) extended this concept by integrating data analytics, enabling predictive waste management and optimized collection routes.

ADVANTAGES:

- Combines monitoring and segregation
- Improves overall system efficiency
- Reduces manual effort

LIMITATIONS:

Increased system complexity

Higher cost compared to basic systems



Fig.2.3 IoT-Based Waste Monitoring System

2.2.4 AI AND MACHINE LEARNING-BASED SYSTEMS

Recent advancements in waste management involve the use of Artificial Intelligence and machine learning techniques.

AI-based systems use image processing and deep learning algorithms to classify waste materials with high accuracy. Cameras capture images of waste, and trained models identify the type of waste (plastic, paper, organic, etc.).

These systems also use predictive analytics to estimate bin fill levels and optimize waste collection schedules.

ADVANTAGES:

- High accuracy in waste classification
- Predictive capabilities
- Scalable for large applications

LIMITATIONS:

- High computational cost
- Requires large datasets for training
- Complex implementation

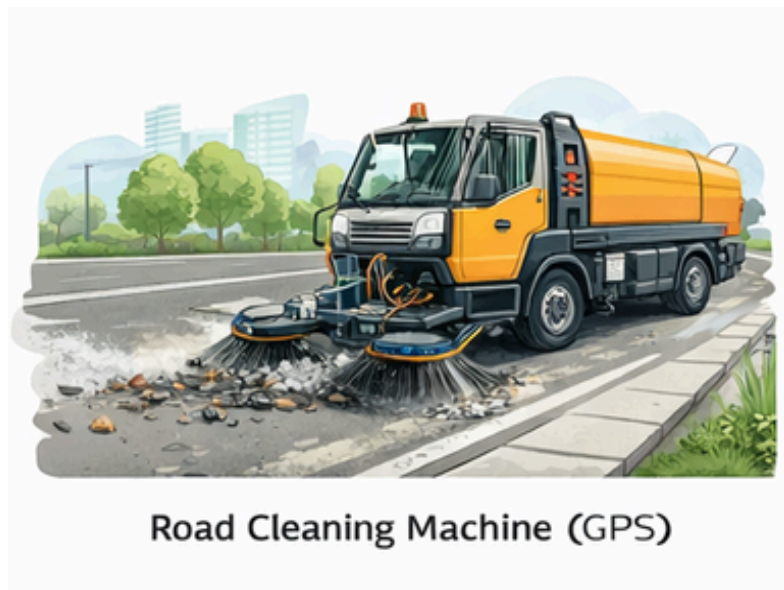


Fig.2.4 Road Cleaning Machine

2.2.5 SMART WASTE MANAGEMENT IN SMART CITIES

Smart city initiatives focus on integrating technology into urban infrastructure to improve efficiency and sustainability. Waste management is a key component of smart city development.

IoT-based smart bins are deployed in public areas to monitor waste levels and send alerts for collection. Data analytics is used to optimize bin placement and collection routes.

Bansal and Sharma (2022) emphasized that smart waste management systems play a crucial role in achieving sustainability goals by reducing pollution and improving resource utilization.

ADVANTAGES:

- Supports sustainable development
- Improves urban cleanliness
- Enables data-driven decision-making

LIMITATIONS:

- Requires large-scale infrastructure
- High implementation cost



Fig.2.5 Garbage Collection Robot

2.2.6 HIGHWAY WASTE MANAGEMENT SYSTEMS

Although many systems focus on urban environments, there is limited research on waste management along highways.

Highway waste management systems require:

- Strategic placement of bins
- Real-time monitoring
- Efficient collection mechanisms

Existing systems do not fully address the unique challenges of highways, such as long distances, irregular waste disposal, and lack of infrastructure.

2.3 COMPARATIVE ANALYSIS

Table 1. comparative analysis of literature survey

Author / Year	System Type	Technology Used	Features	Advantages	Limitations

Patel et al. (2020)	Monitoring + Segregation	IoT + Sensors	Fill level detection, basic segregation	Cost-effective	Limited accuracy
Singh & Yadav (2021)	Smart Monitoring	IoT + Mobile App	Real-time alerts, remote monitoring	Efficient collection	Internet dependent
Bansal & Sharma (2022)	Smart City System	IoT + Cloud	Data analytics, optimization	Scalable, efficient	High cost
AI-Based Systems	Intelligent Segregation	AI + ML	Image classification, prediction	High accuracy	Complex, expensive
Hybrid Systems	Monitoring + Segregation	IoT + Sensors + Cloud	Combined features	Efficient	High complexity

2.4 RESEARCH GAPS

From the above analysis, the following research gaps are identified:

- Lack of integrated systems combining monitoring and segregation effectively
- Limited focus on highway-specific waste management systems
- High cost and complexity of advanced systems
- Dependence on continuous internet connectivity
- Limited scalability for rural and semi-urban areas

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

The existing system represents a **basic smart trash segregation system** designed to automate the process of waste classification using simple electronic components. The primary objective of this system is to reduce manual effort in waste segregation and improve efficiency in handling waste at a small scale.

The system operates using a combination of sensors, a microcontroller, and output devices. It identifies the presence and type of waste using sensors and performs segregation using a mechanical mechanism. Although the system provides automation, it is limited in functionality and does not include advanced features such as real-time monitoring or IoT integration.

3.2 OVERVIEW OF BLOCK DIAGRAM

The block diagram consists of three major functional sections:

1. Input Section

- IR Sensor
- Moisture Sensor
- Power Supply

2. Processing Section

- Arduino Uno Microcontroller

3. Output Section

- Servo Motor (Flap Mechanism)
- LCD Display
- Buzzer

Each section plays a vital role in the operation of the system, forming a complete automation loop from detection to segregation.



Fig.3.1 Overview of Existing Waste Management System

3.3 DETAILED DESCRIPTION OF COMPONENTS

3.3.1 IR SENSOR (INFRARED SENSOR)

The IR sensor is responsible for detecting the presence of waste in the system. It works on the principle of **infrared radiation reflection**.

Working Principle:

- The sensor emits infrared light
- When an object (waste) is placed, the light reflects back
- The receiver detects this reflection
- A signal is sent to the microcontroller

Role in the System:

- Acts as a **triggering mechanism**
- Initiates the entire process only when waste is detected
- Helps in reducing unnecessary operation of the system

Importance:

Without the IR sensor, the system would not be able to identify when waste is inserted, leading to inefficient operation.

3.3.2 MOISTURE SENSOR

The moisture sensor is used to determine the **type of waste** based on its moisture content.

Working Principle:

- Wet waste contains water → conducts electricity
- Dry waste contains little or no moisture → poor conductivity
- Sensor measures electrical conductivity
- Output is sent as an analog or digital signal

Role in the System:

- Classifies waste into:
 - Wet waste (biodegradable)
 - Dry waste (non-biodegradable)

Importance:

This is the **core classification component** of the system. It enables automatic segregation without human intervention.

3.3.3 POWER SUPPLY

The power supply unit provides electrical energy required for the system.

Features:

- Supplies regulated voltage (usually 5V DC)
- Powers sensors, Arduino, and output devices

Role in the System:

- Ensures stable and continuous operation
- Prevents malfunction due to voltage fluctuations

3.3.4 ARDUINO UNO MICROCONTROLLER

The Arduino Uno is the **central processing unit (CPU)** of the system.

Working Principle:

- Receives input signals from sensors
- Processes data using programmed logic
- Sends output signals to control devices

Functions:

- Reads IR sensor data (object detection)
- Reads moisture sensor data (waste classification)
- Executes decision-making logic
- Controls servo motor movement
- Sends data to LCD display
- Activates buzzer

Importance:

The Arduino acts as the **brain of the system**, coordinating all operations and ensuring proper functioning.

3.3.5 SERVO MOTOR (FLAP MECHANISM)

The servo motor is responsible for **physical segregation of waste**.

Working Principle:

- Operates using PWM (Pulse Width Modulation) signals
- Rotates to a specific angle based on input
- Controls a flap or gate mechanism

Role in the System:

- Directs waste into:
 - Wet waste bin
 - Dry waste bin

Importance:

This component converts **electronic decisions into mechanical action**, completing the segregation process.

3.3.6 LCD DISPLAY (STATUS OUTPUT)

The LCD display provides visual feedback to the user.

Functions:

- Displays system messages
- Shows waste classification result
- Indicates system status

Examples of Display Messages:

- “Waste Detected”
- “Wet Waste”
- “Dry Waste”

Importance:

Enhances user interaction and makes the system more informative and user-friendly.

3.3.7 BUZZER (ALERT SYSTEM)

The buzzer is used for **audio indication**.

Working Principle:

- Produces sound when activated
- Controlled by Arduino

Role in the System:

- Alerts when waste is detected
- Indicates system operation

Importance:

Provides immediate feedback, especially useful in noisy or unattended environments.

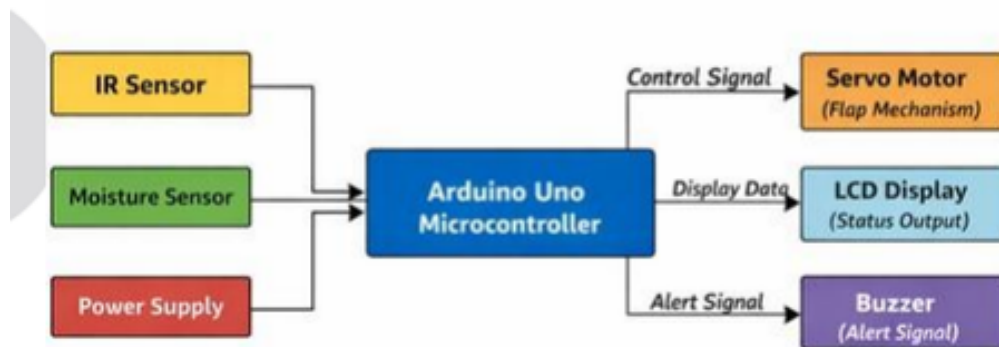


Fig: 3.2 Block Diagram of Existing System

3.4 WORKING OF THE EXISTING SYSTEM

The overall working process is as follows:

1. Waste is inserted into the system
2. IR sensor detects the presence of waste
3. Signal is sent to the Arduino
4. Arduino activates the moisture sensor
5. Moisture sensor analyzes the waste type
6. Data is processed by Arduino
7. Based on classification:
 - Wet waste → servo rotates to wet bin
 - Dry waste → servo rotates to dry bin
8. LCD displays the result
9. Buzzer gives an alert signal

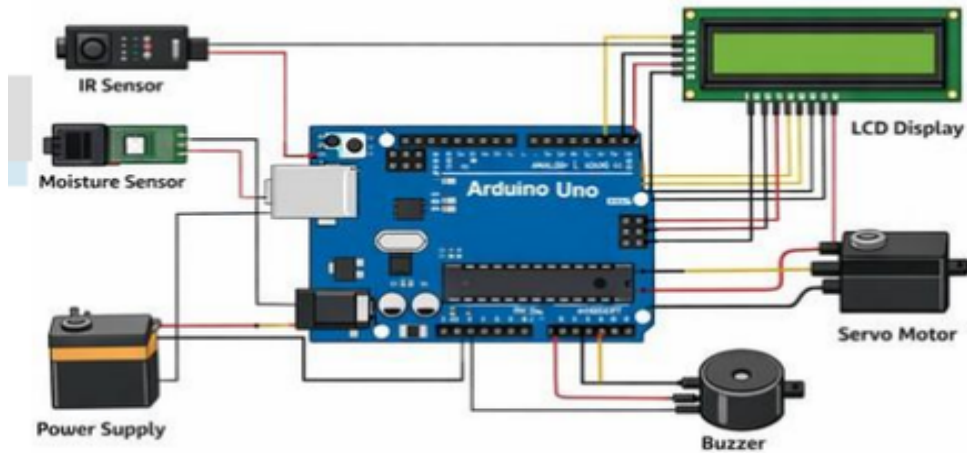


Fig: 3.3 Circuit Diagram of existing system

3.5 ADVANTAGES OF EXISTING SYSTEM

- Simple design and easy implementation
- Low cost and affordable components
- Reduces manual effort
- Provides basic automation
- Suitable for small-scale applications

3.6 LIMITATIONS OF EXISTING SYSTEM

- Limited to only **wet and dry waste classification**
- Sensor accuracy may vary
- No IoT or remote monitoring
- No data storage or analytics
- Not scalable for large systems
- Cannot identify complex waste type

CHAPTER 4

PROPOSED SYSTEM

4.1 EXPLANATION OF PROPOSED SYSTEM

The proposed system is an advanced smart waste management solution that integrates robotics, sensor technology, and communication systems to automate the process of waste collection and segregation. The system is designed using a dual microcontroller architecture consisting of a master Arduino and a slave Arduino, which work together to ensure efficient and

synchronized operation. The master controller is responsible for navigation, obstacle detection, bin monitoring, and communication, while the slave controller manages the robotic arm used for picking and segregating waste. Ultrasonic sensors are used to detect obstacles in front of the robot and to measure the fill level of waste bins. When the robot is in motion, the ultrasonic sensor continuously scans the surroundings to identify the presence of waste or any obstruction. Once an object is detected within a predefined distance, the robot stops, and a command is sent from the master controller to the slave controller to initiate the waste collection process.

The slave Arduino then activates the robotic arm, which consists of multiple servo motors controlling different joints such as the base, shoulder, elbow, gripper, and flipper. The arm moves smoothly towards the detected waste, picks it up using the gripper mechanism, and lifts it to a suitable height. After lifting, the system determines the type of waste using a moisture sensor, which detects the presence of water content in the material. Based on the sensor output, the waste is classified as either wet or dry. The flipper mechanism then rotates to direct the waste into the appropriate bin. Once the waste is successfully deposited, the robotic arm returns to its initial position, and a signal is sent back to the master controller indicating the completion of the task. The robot then resumes its forward movement and continues scanning for additional waste.

In addition to waste collection and segregation, the system also incorporates a bin monitoring mechanism using ultrasonic sensors placed inside the bins. These sensors continuously measure the distance between the sensor and the waste level. When the waste reaches a predefined threshold indicating that the bin is full, the system stops further operation and triggers the GSM module to send an alert message to the concerned authority. This ensures timely waste disposal and prevents overflow, thereby maintaining cleanliness and hygiene. The integration of GSM communication allows remote monitoring and enhances the practicality of the system in real-world applications.

Overall, the proposed system demonstrates a highly efficient and automated approach to waste management by combining mobility, robotic manipulation, sensor-based detection, and communication technologies. It reduces human effort, improves accuracy in waste segregation, and ensures proper waste handling through real-time monitoring and alert mechanisms. This makes the system highly suitable for applications in highways, public areas, and smart city environments.

4.2 WORKING PRINCIPLE

The working principle of the proposed system is based on continuous

coordination between sensors, microcontrollers, and actuators to achieve automated waste collection and segregation. Initially, the robot moves forward autonomously using DC motors controlled by the master Arduino, while an ultrasonic sensor continuously scans the surroundings to detect any object or waste present in its path. When waste is detected within a predefined stopping distance, the robot immediately halts, and the master controller sends a **“PICK”** command to the slave controller through serial communication. Upon receiving this command, the slave Arduino sends an **“ARM_BUSY”** signal back to the master, ensuring that the robot remains stationary during the operation.

The robotic arm then begins its pick-and-place cycle in a sequential manner:

1. The arm moves to the pickup position using coordinated motion of the shoulder and elbow servos.
2. The gripper closes to securely hold the waste object.
3. The arm lifts the waste and moves it to the drop position.
4. A moisture sensor detects the nature of the waste based on its moisture content.
5. Based on the sensor output, the waste is classified as either wet or dry.
6. The flipper mechanism rotates accordingly:
 - Toward the wet bin for wet waste
 - Toward the dry bin for dry waste
7. The waste is dropped into the appropriate bin.
8. The flipper returns to its neutral position.
9. The robotic arm returns to its home position.

After completing the operation, the slave controller sends an **“ARM_DONE”** signal to the master controller, allowing the robot to resume its forward movement and continue scanning for additional waste. This process repeats continuously, enabling efficient and automated waste collection.

At the same time, the system continuously monitors the fill level of both wet and dry bins using ultrasonic sensors. When the waste level reaches a predefined threshold, indicating that the bin is full, the system immediately stops further operation and activates the GSM module to send an alert message to the concerned authority. Once the bin is emptied, the system resets and resumes its operation.

Thus, the proposed system operates in a continuous cycle of detection, collection, segregation, and monitoring, ensuring efficient waste management with minimal human intervention while maintaining cleanliness and preventing overflow.

4.3 FEATURES OF PROPOSED SYSTEM

- Autonomous navigation
- Dual microcontroller system
- Robotic arm automation
- Real-time bin monitoring
- GSM-based alert system
- Smooth servo motion control
- Obstacle detection
- Automatic waste segregation

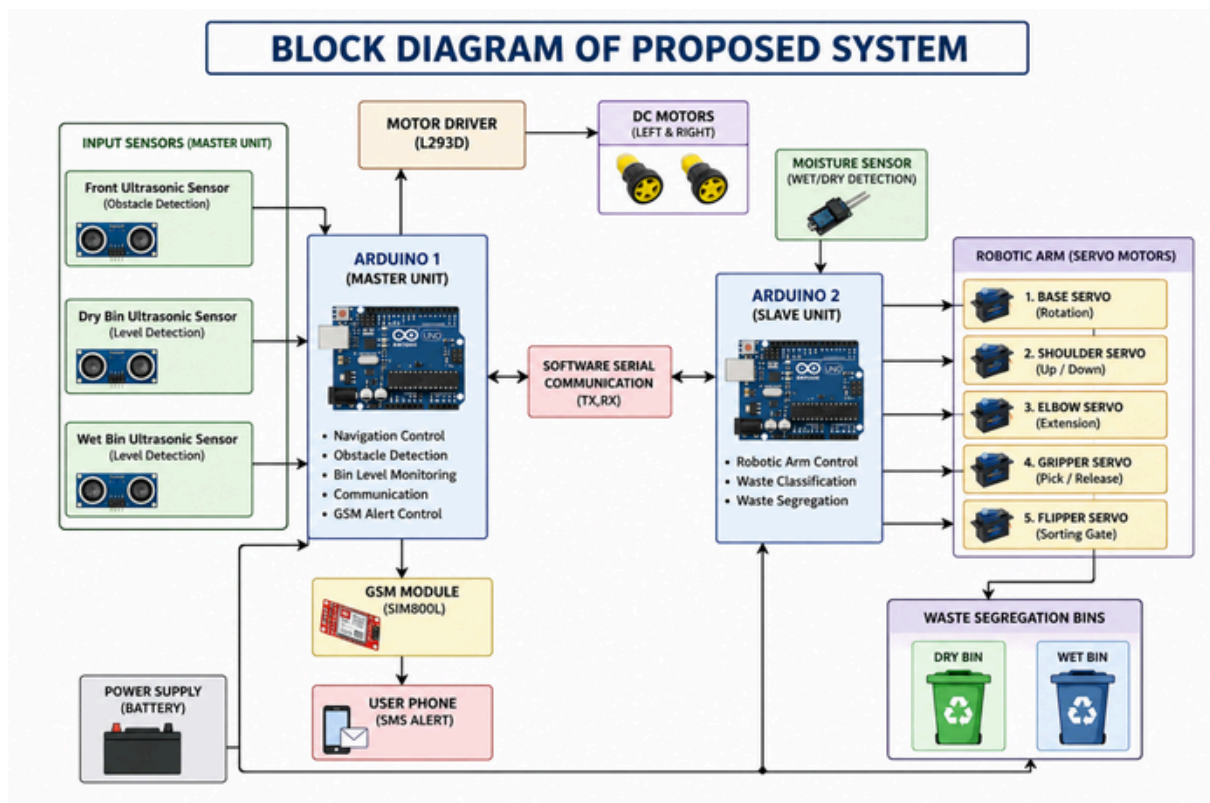


Fig 4.1 block diagram of proposed system

4.4 ADVANTAGES OF PROPOSED SYSTEM

- Reduces human effort and exposure to waste
- Improves efficiency and accuracy
- Prevents bin overflow
- Suitable for highways and smart cities
- Cost-effective compared to industrial systems
- Scalable and upgradeable
- Real-time operation

CHAPTER 5

TOOLS REQUIRED

5.1 TOOLS AND TECHNOLOGIES EMPLOYED

The tools and technologies employed in this system include both hardware and interfacing components that work together to build the Smart Trash Bot application. The Arduino Uno microcontroller acts as the main processing unit and controls the overall operation of the system. A dual Arduino architecture is used in this project to divide tasks efficiently, where one controller manages movement, detection, and communication, while the other handles robotic arm operations and waste segregation.

Servo motors are used for mechanical movement to perform robotic arm actions such as picking and placing waste materials. DC motors are used for the movement of the robot along the roadside. The ultrasonic sensor is used for detecting waste objects as well as monitoring the bin level. The wet sensor is used to detect moisture content in waste, which helps in classifying waste into wet and dry categories.

An LCD display is used to display system status, messages, and alerts, improving user interaction with the system. A GSM module is used for communication and sends alert messages when the waste bin becomes full. A buzzer is included to provide audible alerts during system events such as bin full indication or error conditions.

A DC-DC converter is used to regulate the voltage supply and ensure stable operation of all electronic components. Jumper wires are used to establish proper electrical connections between all components, and a breadboard is used for easy prototyping and circuit assembly without soldering. The power supply or battery provides the necessary electrical power to operate the entire system.

5.2 HARDWARE TOOLS

The hardware assembly of the Smart Trash Bot system involves interfacing multiple components with Arduino microcontrollers to achieve automated waste collection and segregation. The Arduino Uno acts as the central controller, managing communication between sensors, actuators, and communication modules.

An ultrasonic sensor is connected to detect waste objects and monitor bin levels. A wet sensor is used to identify the type of waste based on moisture content. Servo motors are used for controlling the robotic arm and waste segregation mechanism. A robotic arm kit is used to pick and place waste materials.

DC motors are used for robot movement, and a motor driver is used to control their speed and direction. The GSM module is connected to the Arduino to send alert messages when the bin is full. An LCD display is interfaced to show system messages and status updates. A buzzer is used to provide audio alerts. The system is powered using a battery or regulated power supply.

5.2.1 ARDUINO UNO MICROCONTROLLER

The Arduino Uno is a widely used microcontroller board based on the ATmega328P and is commonly used in embedded system applications. It operates at a voltage of 5V and provides multiple digital and analog input/output pins for interfacing with sensors, actuators, and communication modules.

In this project, two Arduino Uno boards are used to implement a dual-controller system. One Arduino acts as the master controller, which manages robot movement, waste detection, bin level monitoring, and GSM communication. The second Arduino acts as the slave controller, which is responsible for controlling the robotic arm and performing waste segregation.

The Arduino Uno supports various communication protocols such as UART, SPI, and I2C, which makes it highly flexible for connecting different modules. It processes input signals from sensors and generates output signals to control motors, servos, and other devices.

The Arduino is programmed using the Arduino IDE with embedded C/C++ language. Its simplicity, flexibility, and wide support make it suitable for real-time embedded applications.



Fig.5.1 Arduino Uno Microcontroller

- Based on ATmega328P microcontroller
- Operates at 5V supply
- Provides digital and analog input/output pins
- Supports UART, SPI, and I2C communication
- Used as master and slave controllers
- Controls sensors, motors, and communication modules
- Easy to program using Arduino IDE

5.2.2 SERVO MOTOR CONTROL

The servo motor is used for controlling the robotic arm and waste segregation mechanism in the Smart Trash Bot system. It is controlled by the Arduino microcontroller through software programming. A servo motor is a type of motor that can rotate to a specific angle based on input signals. In this project, it is programmed to move the robotic arm joints and also direct waste into

specific bins based on classification.

The Arduino sends PWM (Pulse Width Modulation) signals to control the rotation angle of the servo motor. The servo rotates to different positions to perform operations such as picking, lifting, and placing waste. When waste is detected, the servo motors move the robotic arm towards the object, pick it, and place it into the segregation unit. Based on classification, another servo directs the waste into wet or dry bins.



Fig.5.2 Servo Motor

- Provides precise angular movement
- Operates using PWM signals
- Used in robotic arm joints
- Used for waste segregation
- Ensures accurate positioning
- Low power consumption
- High efficiency and reliability

5.2.3 WET SENSOR

The wet sensor is used to detect moisture content in waste materials. It plays a key role in the classification process of the Smart Trash Bot system. The sensor works based on electrical conductivity. It consists of probes that detect the presence of water. When moisture is present, conductivity increases and the sensor outputs a signal indicating wet waste. The Arduino reads this signal and

classifies the waste accordingly. If moisture is detected, the waste is categorized as wet waste; otherwise, it is considered dry waste.

This method improves waste segregation and helps in proper recycling and disposal.



Fig.5.3 Wet Sensor

- Detects moisture content
- Works on conductivity principle
- Used for waste classification
- Simple and effective
- Low cost and easy to use
- Improves segregation accuracy

5.2.4 ROBOTIC ARM KIT

The robotic arm kit is used for picking and placing waste materials in the Smart Trash Bot system. It consists of multiple joints controlled by servo motors. The arm moves towards the detected waste, grips it using a gripper mechanism, and transfers it to the segregation unit. After placing the waste, it returns to its original position.

The robotic arm enables automation and reduces manual effort in waste collection.

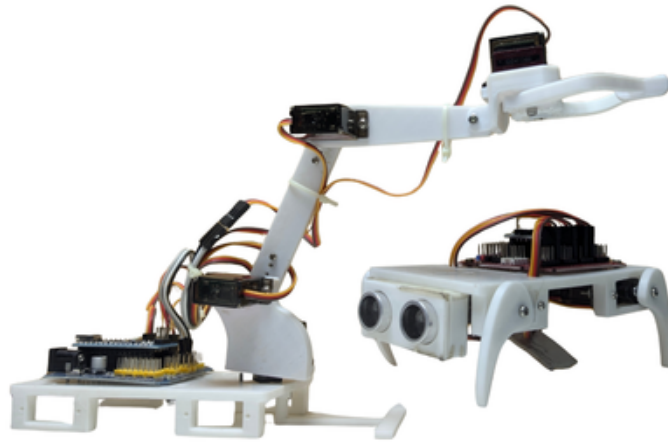


Fig.5.4 Robotic Arm Kit

- Multi-joint mechanical structure
- Controlled using servo motors
- Performs pick and place operations
- Reduces human intervention
- Improves efficiency and automation
- Enables precise movement

5.2.5 ULTRASONIC SENSOR

The ultrasonic sensor is used to detect waste objects and measure the bin level. It works by sending ultrasonic waves and measuring the time taken for the echo to return. The distance is calculated based on the time delay of the reflected signal. This helps in identifying the presence of waste and determining how full the bin is. When waste is detected within a certain range, the system activates the robotic arm. When the bin is full, the system sends an alert through the GSM module.



Fig.5.5 Ultrasonic Sensor

- Measures distance using sound waves
- Used for object detection
- Used for bin level monitoring
- High accuracy and reliability
- Easy to interface with Arduino
- Widely used in robotics

5.2.6 LCD DISPLAY

The LCD display is used to provide visual feedback to the user by displaying system messages and instructions. It helps users understand the working status of the system. The display shows messages such as system ready, waste detected, bin full, and alert notifications. It improves system usability and monitoring.



Fig.5.6 LCD Display

- Displays system status
- Shows alerts and messages
- Improves user interaction
- Low power consumption
- Easy to interface

5.2.7 DC-DC CONVERTER

The DC-DC converter is used to regulate voltage levels for different components in the system. It ensures that all components receive the required voltage for proper functioning.

It protects components from voltage fluctuations and improves system reliability.



Fig.5.7 DC-DC Converter

- Regulates voltage supply
- Ensures stable operation
- Protects components
- Improves efficiency

5.2.8 BUZZER

The buzzer is used to provide audio alerts in the system. It produces sound when triggered by the Arduino.

It is used to notify events such as bin full alerts or system errors.



Fig.5.8 Buzzer

- Provides audio alerts
- Simple and effective
- Low power consumption
- Used for notifications

5.2.9 GSM MODULE

The GSM module is used for communication in the Smart Trash Bot system. It sends SMS alerts when the waste bin becomes full.

It works using a SIM card and communicates over the cellular network.



Fig.5.9 GSM Module

- Sends real-time SMS alerts
- Uses SIM network
- Enables remote communication
- Improves monitoring system

5.2.10 MOTOR DRIVER

The motor driver is used to control DC motors. It acts as an interface between the Arduino and the motors.

It allows control of motor speed and direction.



Fig.5.10 Motor Driver

- Controls motor speed
- Controls direction
- Handles high current
- Protects microcontroller

5.2.11 DC MOTOR

DC motors are used for the movement of the robot. They allow the robot to move along the roadside path.

They are controlled using the motor driver and Arduino.



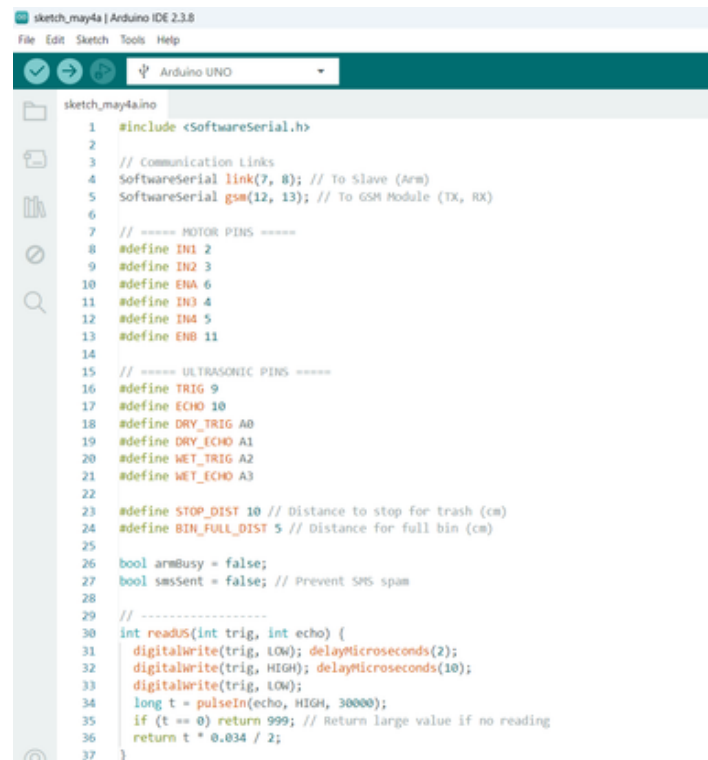
Fig.5.11 DC Motor

- Provides movement
- Easy to control
- Cost-effective
- Widely used in robotics

5.3 SOFTWARE TOOL

5.3.1 ARDUINO IDE

The Arduino Integrated Development Environment (IDE) is an open-source software platform used for writing, compiling, and uploading code to Arduino microcontrollers. It supports programming in C and C++ languages. It provides features such as a text editor, compiler, and serial monitor, which help in debugging and testing the system. It also supports integration of various libraries required for hardware interfacing.



```

sketch_may4a | Arduino IDE 2.3.8
File Edit Sketch Tools Help

sketch_may4a.ino
1 #include <SoftwareSerial.h>
2
3 // Communication links
4 SoftwareSerial link(7, 8); // To Slave (Arm)
5 SoftwareSerial gsm(12, 13); // To GSM Module (TX, RX)
6
7 // ===== MOTOR PINS =====
8 #define IN1 2
9 #define IN2 3
10 #define ENA 6
11 #define IN3 4
12 #define IN4 5
13 #define ENB 11
14
15 // ===== ULTRASONIC PINS =====
16 #define TRIG 9
17 #define ECHO 10
18 #define DRV_TRIG A0
19 #define DRV_ECHO A1
20 #define MET_TRIG A2
21 #define MET_ECHO A3
22
23 #define STOP_DIST 10 // Distance to stop for trash (cm)
24 #define BIN_FULL_DIST 5 // Distance for full bin (cm)
25
26 bool armBusy = false;
27 bool smsSent = false; // Prevent SMS spam
28
29 // -----
30 int readUS(int trig, int echo) {
31     digitalWrite(trig, LOW); delayMicroseconds(2);
32     digitalWrite(trig, HIGH); delayMicroseconds(10);
33     digitalWrite(trig, LOW);
34     long t = pulseIn(echo, HIGH, 30000);
35     if (t == 0) return 999; // Return large value if no reading
36     return t * 0.034 / 2;
37 }

```

Fig.5.12 Arduino IDE

- Easy to use and beginner-friendly
- Open-source software
- Supports multiple libraries
- Provides serial monitor for debugging
- Compatible with multiple platforms

5.3.2 ARDUINO LIBRARIES

Libraries are pre-written codes that simplify programming. They help in interfacing sensors, motors, and communication modules.

Libraries reduce coding complexity and improve development efficiency.

- Reduces coding complexity
- Saves development time
- Improves reliability
- Easy hardware integration
- Provides ready-made functions

CHAPTER 6

RESULTS AND DISCUSSION

6.1 DESIGN AND IMPLEMENTATION OF PROPOSED SYSTEM

The proposed **Smart Trash Bot: Autonomous Waste Collection and Segregation System with GSM Module** has been successfully designed and implemented using embedded system technology. The system integrates multiple sensors, actuators, and communication modules to automate waste collection, detection, and segregation.

The design consists of a mobile robotic platform equipped with DC motors for movement, controlled using a motor driver module. The Arduino Uno microcontroller acts as the central processing unit, coordinating all system operations including navigation, waste detection, segregation, and communication.

The ultrasonic sensor is used to detect obstacles and identify waste presence along the roadside. It measures the distance between the robot and objects, allowing the robot to stop when waste is detected. The wet sensor is used to identify the type of waste by detecting moisture content, enabling classification into wet and dry waste.

A robotic arm (4-DOF) is mounted on the robot to pick and place waste into appropriate bins. The arm is controlled using servo motors, which provide precise angular movement required for accurate waste handling.

The system also includes a GSM module for communication. When the waste bin reaches a certain threshold level, the system automatically sends an alert message to the concerned authorities.

The LCD display provides real-time feedback such as system status, detection messages, and alerts. A buzzer is used to indicate system actions and warnings.

Key design features:

- Autonomous movement using DC motors
- Waste detection using ultrasonic sensor
- Waste classification using wet sensor
- Robotic arm for automated collection
- GSM-based alert system
- Real-time monitoring using LCD display

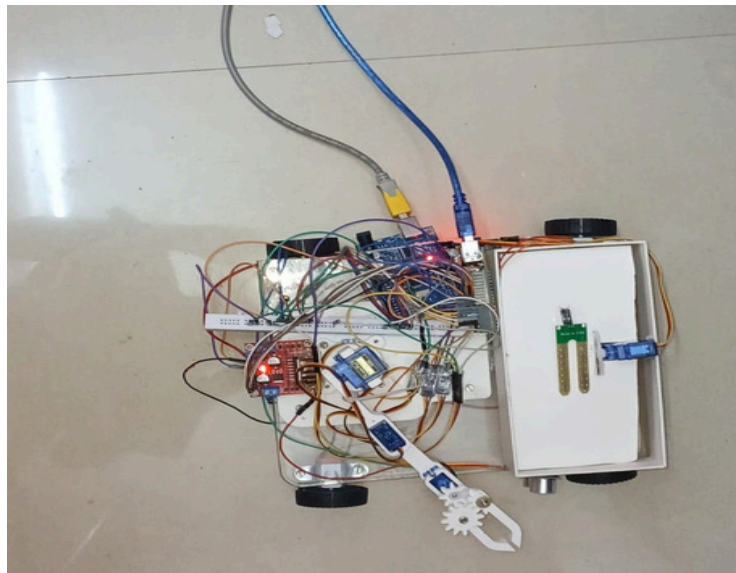


Fig.6.1 Design of Proposed System View

6.2 ACTUATION AND OUTPUT MECHANISM

The actuation mechanism plays a vital role in converting electrical signals into physical actions within the system. In the Smart Trash Bot, actuators such as DC motors and servo motors are used to perform movement and waste handling operations.

DC motors are responsible for the movement of the robot. These motors are controlled using the L298N motor driver, which allows forward, backward, left, and right motion. The Arduino sends control signals to the motor driver, enabling navigation along the roadside.

The robotic arm is controlled using multiple servo motors. These motors

operate based on Pulse Width Modulation (PWM) signals generated by the Arduino Uno. Each servo controls a specific joint of the robotic arm, enabling movements such as lifting, rotating, and placing waste.

The system follows a sequence of operations:

- Robot moves forward along the path
- Ultrasonic sensor detects waste
- Robot stops near waste
- Robotic arm picks the waste
- Wet sensor classifies waste
- Waste is placed into appropriate bin

The LCD display acts as a visual output device, showing messages such as:

- “Waste Detected”
- “Picking Waste”
- “Wet Waste” / “Dry Waste”
- “Bin Full”

The buzzer provides audio alerts for:

- Waste detection
- Bin full condition
- System errors

The GSM module acts as a communication output, sending SMS alerts when the bin is full.

This coordinated actuation and output mechanism ensures:

- Accurate waste handling
- Real-time feedback
- Efficient automation
- Improved user interaction



Fig.6.2 Implementation of Proposed System (Waste Detection)

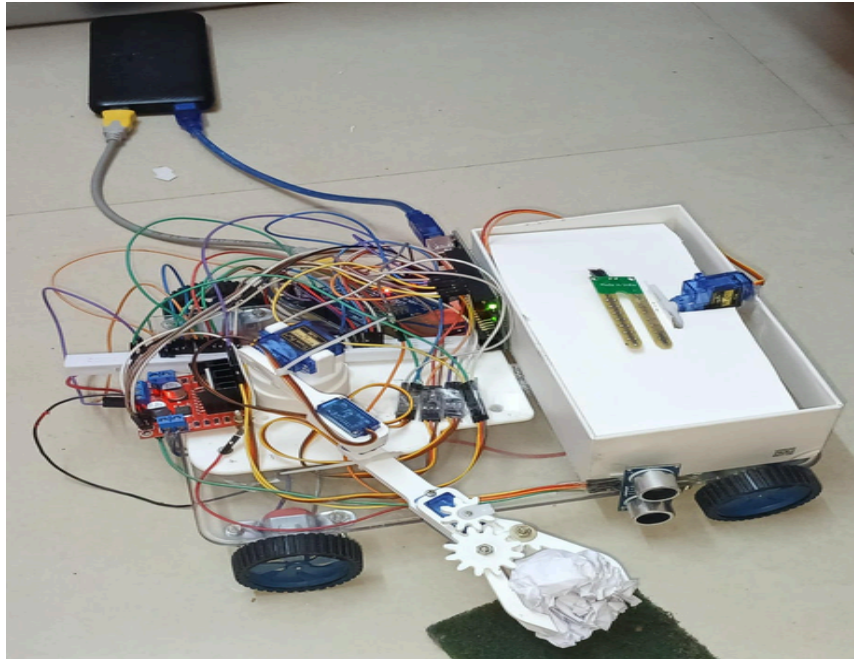


Fig.6.3 Implementation of Proposed System (Waste Picking)

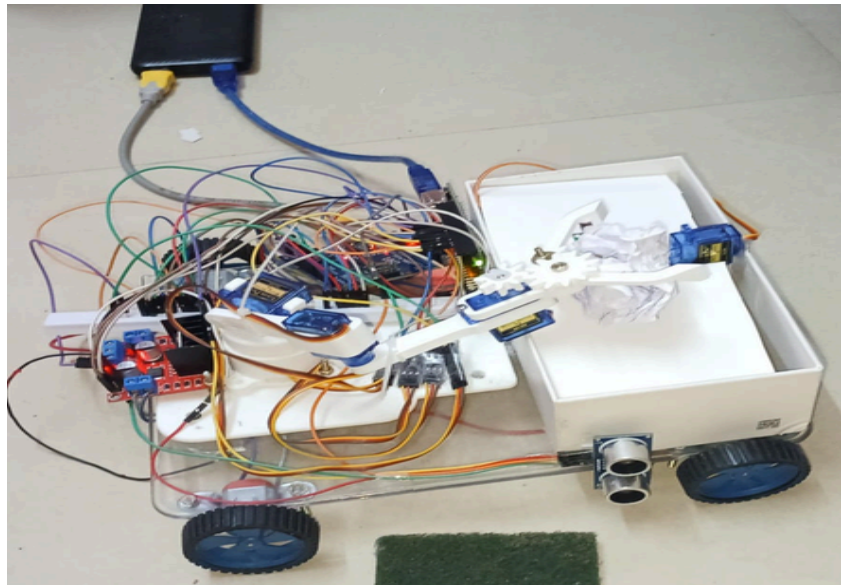


Fig.6.4 Implementation of Proposed System (Waste Dropping)

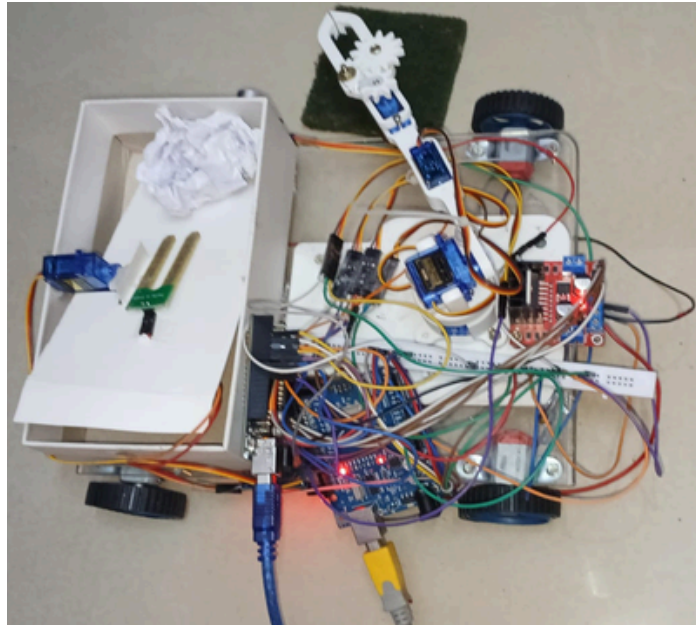


Fig.6.5 Implementation of Proposed System (Waste Separation)

6.3 COMPARISON BETWEEN EXISTING SYSTEM AND PROPOSED SYSTEM

Table 6.1 Comparison Between Existing System and Proposed System

S.NO	ASPECT	EXISTING SYSTEM	PROPOSED SYSTEM
1	System Type	Manual waste collection	Autonomous robotic system
2	Control Method	Human-operated	Arduino-based automated control
3	Waste Detection	Manual observation	Ultrasonic sensor-based detection
4	Waste Segregation	Not available	Automatic wet/dry segregation
5	Safety	Risky for workers	Safe, minimal human

			involvement
6	Efficiency	Low	High efficiency
7	Monitoring	No monitoring system	Real-time monitoring
8	Alert System	Not available	GSM-based alert system
9	Automation Level	Very low	Fully automated
10	Labor Requirement	High	Low
11	Environmental Impact	High pollution	Reduced pollution
12	Response Time	Slow	Fast response
13	Smart Features	Not available	IoT-enabled features
14	Waste Handling	Manual	Robotic arm handling
15	Data Processing	Not available	Real-time processing
16	Reliability	Moderate	High reliability
17	Maintenance	Frequent	Moderate
18	Cost	Low initial cost	Moderate but cost-effective
19	Scalability	Difficult	Easily scalable

20	Application Areas	Basic cleaning	Smart cities, highways
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CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The Smart Trash Bot: Autonomous Waste Collection and Segregation System with GSM Module presents an innovative and efficient solution to modern waste management challenges. The system successfully integrates embedded systems, sensors, and automation to reduce human effort and improve waste handling efficiency. The use of the Arduino Uno microcontroller enables effective coordination between various components such as sensors, motors, and communication modules. The ultrasonic sensor ensures accurate waste detection, while the wet sensor enables proper waste classification. The robotic arm enhances automation by enabling precise pick-and-place operations. The integration of GSM technology allows real-time communication by sending alerts when waste bins are full, ensuring timely waste disposal. The system significantly reduces manual labor, enhances safety, and minimizes environmental pollution. It is cost-effective and can be implemented in highways, smart cities, and urban environments. Overall, this project demonstrates how automation and IoT can be applied to solve real-world problems effectively. It serves as a strong foundation for developing advanced intelligent waste management systems.

7.2 FUTURE WORK

The proposed system can be further enhanced by integrating advanced

technologies to improve performance, intelligence, and scalability.

Future improvements include:

- **AI-based Waste Recognition** Using image processing and machine learning to identify different types of waste.
- **GPS Integration** Tracking robot location for better navigation and monitoring.
- **Mobile Application** Developing an app for real-time monitoring and control.
- **Solar Power Integration** Using solar panels for energy-efficient operation.
- **Camera Module** Capturing images for monitoring and analysis.
- **Cloud Integration** Storing data for analytics and smart decision-making.
- **Advanced Sensors** Using gas sensors to detect hazardous waste.
- **Improved Robotic Arm** Enhancing payload capacity and flexibility.
- **Swarm Robotics** Using multiple robots for large-scale cleaning.
- **Automatic Charging System** Self-charging mechanism for continuous operation.

These improvements can transform the system into a fully intelligent and scalable smart waste management solution.

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APPENDIX

APPENDIX A – CIRCUIT DIAGRAM DESCRIPTION

The system consists of the following connections:

- Arduino Uno connected to motor driver (L298N)
- DC motors connected to motor driver
- Ultrasonic sensor connected to digital pins
- Wet sensor connected to analog pin
- Servo motors connected to PWM pins
- GSM module connected via serial communication
- LCD connected using I2C interface
- Power supply through DC-DC converter

APPENDIX B – WORKING ALGORITHM

1. Start system
2. Initialize all components
3. Move robot forward
4. Detect waste using ultrasonic sensor
5. Stop robot
6. Activate robotic arm
7. Detect waste type using wet sensor

8. Segregate waste into bins
9. Monitor bin level
10. If bin full → send SMS via GSM
11. Display status on LCD
12. Repeat process

APPENDIX C – SAMPLE OUTPUT MESSAGES

• “System Initialized”• “Waste Detected”• “Picking Waste”• “Wet Waste Detected”• “Dry Waste Detected”• “Bin Full – Alert Sent”

APPENDIX D – APPLICATION AREAS

• Smart cities• Highways• Industrial areas• Public places• Railway stations• Airports

PROGRAM:

ARDUINO 1:

```
#include <SoftwareSerial.h>
```

```
// Communication Links
```

```
SoftwareSerial link(7, 8); // To Slave (Arm)
```

```
SoftwareSerial gsm(12, 13); // To GSM Module (TX, RX)
```

```
// ===== MOTOR PINS =====
```

```
#define IN1 2
```

```
#define IN2 3
```

```
#define ENA 6
```

```

#define IN3 4
#define IN4 5
#define ENB 11

// ===== ULTRASONIC PINS =====
#define TRIG 9
#define ECHO 10
#define DRY_TRIG A0
#define DRY_ECHO A1
#define WET_TRIG A2
#define WET_ECHO A3

#define STOP_DIST 10 // Distance to stop for trash (cm)
#define BIN_FULL_DIST 5 // Distance for full bin (cm)

bool armBusy = false;
bool smsSent = false; // Prevent SMS spam

// -----
int readUS(int trig, int echo) {
    digitalWrite(trig, LOW); delayMicroseconds(2);
    digitalWrite(trig, HIGH); delayMicroseconds(10);
    digitalWrite(trig, LOW);
    long t = pulseIn(echo, HIGH, 30000);
    if (t == 0) return 999; // Return large value if no reading
    return t * 0.034 / 2;
}

void moveForward() {
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
    analogWrite(ENA, 180); analogWrite(ENB, 180);
}

void stopMotors() {
    digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
    analogWrite(ENA, 0); analogWrite(ENB, 0);
}

```

```
}
```

```
void sendSMS(String message) {  
    gsm.println("AT+CMGF=1"); // Set SMS to text mode  
    delay(1000);  
    gsm.println("AT+CMGS=\"+1234567890\""); // REPLACE WITH YOUR NUMBER  
    delay(1000);  
    gsm.print(message);  
    delay(100);  
    gsm.write(26); // ASCII code for CTRL+Z to send  
    delay(5000);  
}
```

```
void setup() {  
    Serial.begin(9600);  
    link.begin(9600);  
    gsm.begin(9600);  
  
    pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT); pinMode(ENA, OUTPUT);  
    pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT); pinMode(ENB, OUTPUT);  
    pinMode(TRIG, OUTPUT); pinMode(ECHO, INPUT);  
    pinMode(DRY_TRIG, OUTPUT); pinMode(DRY_ECHO, INPUT);  
    pinMode(WET_TRIG, OUTPUT); pinMode(WET_ECHO, INPUT);  
  
    stopMotors();  
    Serial.println("MASTER READY");  
}
```

```
void loop() {  
    // 1. Check for updates from Arm  
    if (link.available()) {  
        String msg = link.readStringUntil('\n');  
        msg.trim();  
        if (msg == "ARM_BUSY") armBusy = true;  
        if (msg == "ARM_DONE") armBusy = false;  
    }  
  
    // 2. Read Sensors
```

```
int obs = readUS(TRIG, ECHO);
int dryDist = readUS(DRY_TRIG, DRY_ECHO);
int wetDist = readUS(WET_TRIG, WET_ECHO);
```

```
bool dryFull = (dryDist<= BIN_FULL_DIST);
bool wetFull = (wetDist<= BIN_FULL_DIST);
```

```
// 3. Handle Bin Full (GSM)
```

```
if (dryFull || wetFull) {
    stopMotors();
    if (!smsSent) {
        Serial.println("BIN FULL! Sending SMS...");
        sendSMS("Alert: Waste bin is full. Please empty.");
        smsSent = true;
    }
    return;
} else {
    smsSent = false; // Reset when emptied
}
```

```
// 4. Handle Arm Activity
```

```
if (armBusy) {
    stopMotors();
    return;
}
```

```
// 5. Navigation Logic
```

```
if (obs<= STOP_DIST) {
    stopMotors();
    Serial.println("OBSTACLE DETECTED - TRIGGERING ARM");
    link.println("PICK"); // Only trigger if something is in front
    delay(1000); // Debounce trigger
} else {
    moveForward();
}

delay(100);
}
```

ARDUINO 2:

```
#include <Servo.h>
```

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial link(7, 8); // RX, TX to Master
```

```
Servo base, shoulder, elbow, gripper, flipper;
```

```
// ===== PINS =====
```

```
#define BASE_PIN 5
```

```
#define SHOULDER_PIN 6
```

```
#define ELBOW_PIN 9
```

```
#define GRIPPER_PIN 10
```

```
#define FLIPPER_PIN 3
```

```
#define WET_SENSOR_PIN 4
```

```
// ===== POSITIONS =====
```

```
#define BASE_HOME 90
```

```
#define FLIP_CENTER 110
```

```
#define FLIP_LEFT 60 // Dry
```

```
#define FLIP_RIGHT 180 // Wet
```

```
#define SHOULDER_HOME 90
```

```
#define SHOULDER_PICK 20
```

```
#define SHOULDER_DROP 90
```

```
#define ELBOW_HOME 100
```

```
#define ELBOW_PICK 100
```

```
#define ELBOW_DROP 0
```

```
#define GRIPPER_OPEN 180
```

```
#define GRIPPER_CLOSE 20
```

```
// ===== STATE TRACKING =====
```

```
int sPos = SHOULDER_HOME;
```

```
int ePos = ELBOW_HOME;
```

```
int fPos = FLIP_CENTER;
```

```
// ===== MOVE ARM FUNCTION =====
void moveBoth(int sTarget, int eTarget, int speedDelay = 15) {
    while (sPos != sTarget || ePos != eTarget) {
        if (sPos < sTarget) sPos++;
        else if (sPos > sTarget) sPos--;

        if (ePos < eTarget) ePos++;
        else if (ePos > eTarget) ePos--;

        shoulder.write(sPos);
        elbow.write(ePos);
        delay(speedDelay);
    }
}
```

```
// ===== MOVE FLIPPER FUNCTION =====
void moveFlipper(int target, int speedDelay = 10) {
    while (fPos != target) {
        if (fPos < target) fPos++;
        else if (fPos > target) fPos--;
        flipper.write(fPos);
        delay(speedDelay);
    }
}
```

```
void setup() {
    Serial.begin(9600);
    link.begin(9600); // Communicate with Master

    base.attach(BASE_PIN);
    shoulder.attach(SHOULDER_PIN);
    elbow.attach(ELBOW_PIN);
    gripper.attach(GRIPPER_PIN);
    flipper.attach(FLIPPER_PIN);

    pinMode(WET_SENSOR_PIN, INPUT);
}
```

```

// Initialize positions
base.write(BASE_HOME);
shoulder.write(sPos);
elbow.write(ePos);
gripper.write(GRIPPER_OPEN);
flipper.write(fPos);

Serial.println("ARM SLAVE READY");
}

void loop() {
  // Check if Master sent a "PICK" command
  if (link.available()) {
    String cmd = link.readStringUntil('\n');
    cmd.trim();

    if (cmd == "PICK") {
      // 1. Tell Master to stop moving motors
      link.println("ARM_BUSY");
      Serial.println("Starting Cycle...");

      // 2. MOVE TO PICK
      moveBoth(SHOULDER_PICK, ELBOW_PICK);
      delay(500);

      // 3. CLOSE GRIPPER
      gripper.write(GRIPPER_CLOSE);
      delay(1000);

      // 4. MOVE TO DROP
      moveBoth(SHOULDER_DROP, ELBOW_DROP);
      delay(500);

      // 5. OPEN GRIPPER
      gripper.write(GRIPPER_OPEN);
      delay(1000);

      // 6. SORTING

```

```
if (digitalRead(WET_SENSOR_PIN) == HIGH) {  
    moveFlipper(FLIP_RIGHT); // Wet  
} else {  
    moveFlipper(FLIP_LEFT); // Dry  
}  
delay(1500);  
moveFlipper(FLIP_CENTER);  
  
// 7. RETURN HOME  
moveBoth(SHOULDER_HOME, ELBOW_HOME);  
  
// 8. Tell Master it can move again  
link.println("ARM_DONE");  
Serial.println("Cycle Complete.");  
}  
}  
}
```